Temporal Clustering for Pancreatic Cancer: A Research Proposal

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Background and Motivation

Pancreatic cancer remains one of the deadliest cancers, with a five-year survival rate below 12 percent. Early-stage detection could dramatically improve outcomes, as patients diagnosed with localized disease have five-year survival rates approaching 40%, compared to less than 3% for those with metastatic disease [1]. Despite this, early diagnosis remains elusive because pancreatic cancer is often asymptomatic until late stages and currently lacks cost-effective population-level screening tools.

Placido et al. recently demonstrated that sequential deep-learning models, specifically gated recurrent units (GRU) and Transformer architectures, trained on longitudinal disease trajectories from more than nine million patients can predict pancreatic cancer risk with an area under the receiver operating characteristic (AUROC) curve of 0.88 up to 36 months before diagnosis [2]. Their work showed that temporal information embedded in electronic health records (EHRs) holds significant predictive value.

However, several critical limitations identified in their work prevent clinical implementation:

- **Limited discovery of disease-progression subtypes:** CancerRiskNet treats all trajectories as deriving from a single generative process. The model's latent space is optimized for risk prediction but does not explicitly separate distinct temporal pathways (e.g., metabolic, inflammatory, or rapid-onset patterns). Without such stratification, clinicians cannot tailor surveillance to subgroup-specific natural history.
- **Modest cross-system generalizability:** When the Denmark-trained model was applied to U.S. Veterans Affairs data, AUROC fell from 0.88 to 0.71 [2], underscoring substantial heterogeneity in coding practices, health-care utilization, and population characteristics. This indicates that current approaches may not generalize well across different healthcare systems.
- **Absence of actionable risk stratification:** Even when models identify high-risk patients, they don't provide clear guidance on appropriate clinical actions. The authors note the need for a practical scenario for surveillance programs.

This research proposal addresses these gaps by augmenting Placido et al.'s encoder with deep embedded clustering (DEC) to discover progression phenotypes, validating cluster reproducibility across diverse health systems, and translating clusters into clinically actionable risk-stratification guidelines.

Research Program

Discover temporal disease trajectories phenotypes with Deep Embedded Clustering

Relevant Background

Conventional clustering algorithms struggle with the irregular sampling, varying trajectory lengths, and high dimensionality of EHR sequences. Recent advances in temporal representation learning offer two complementary approaches. First, sequence encoders such as GRUs and Transformers capture long-range dependencies and event ordering, achieving state-of-the-art performance in cancer risk prediction [2]. Second, Deep Embedded Clustering (DEC) is an unsupervised learning technique that jointly learns a latent representation and clustering structure by minimizing the Kullback–Leibler (KL) divergence between soft cluster assignments and a target distribution. This process iteratively updates both the encoder network and the cluster centroids, gradually sharpening the confidence of assignments and yielding well-separated clusters without requiring labels [4]. An extension, Improved Deep Embedded Clustering (IDEC), augments this framework with a reconstruction loss from an autoencoder, ensuring that the learned embedding space retains fine-grained local structure. This combination allows IDEC to preserve trajectory-specific patterns while still discovering global phenotypic groupings [5]. These methods have proven effective in phenotyping acute-care trajectories [6], but have not yet been applied to pancreatic cancer progression.

Methods

- **Data preprocessing:** We will reproduce Placido et al.'s pipeline: convert ICD-10 (and mapped ICD-9) codes to three-character granularity. Truncate trajectories at least 12 months before diagnosis for cases.
- **Unsupervised encoder training for clustering**: We train a GRU / Transformer sequence-to-sequence auto-encoder to reconstruct each patient's ICD-timestamp trajectory. The architecture re-uses the same diagnosis and time-gap embeddings described by Placido et al., retaining architectural continuity while avoiding any risk-prediction bias. The bottleneck vector captures global trajectory without depending on the outcome.
- **Deep Embedded Clustering (DEC/IDEC):** The auto-encoder embeddings are fed into DEC: we run k-means once to seed cluster centroids, then alternate between computing soft membership probabilities and jointly updating centroids and encoder weights by minimising KL-divergence. In the IDEC variant we keep a reconstruction loss so local detail is preserved. This optimization creates clusters that make sense overall, by pulling similar cases closer together, and also reflect local details, by keeping the important patterns in the timeline. As a result, the phenotypes represent real patterns in the data.
- **Interpretability and clinical anchoring:** After each DEC run we will summarise every cluster, top ICD-3 codes and median event-timeline derived via integrated gradients, then present it to a oncology panel. A cluster is retained only if reviewers agree it matches a known or plausible pathogenic pathway, shows a coherent sentinel-to-cancer timeline, and covers ≥ 1 % of patients. Otherwise it is merged or the model is re-tuned. This keeps clusters both technically interpretable and clinically useful.

- **Internal validation:** Optimal K will be selected via the elbow method on Calinski–Harabasz scores. Cluster quality will be measured by silhouette (>0.6) and Davies–Bouldin (<1.0). We will compare DEC, IDEC, and a baseline k-means on pre-trained embeddings.

Expected Results
We expect to identify 5–10 clinically meaningful clusters that capture distinct progression pathways, such as metabolic (e.g., obesity, type 2 diabetes) or inflammatory (e.g., recurrent pancreatitis) patterns. These clusters should differ in both diagnostic composition and timing of key events. Integrated gradient analyses will help validate known risk factors and may reveal novel early indicators. The resulting phenotypes will support more personalized surveillance strategies and could inform future research into the biological mechanisms of pancreatic cancer progression.

Validate cluster robustness and model generalizability across health systems

Relevant Background

Placido et al. showed a sharp performance decline when their Danish-trained model was applied to U.S. VA data, attributing the drop to differences in trajectory length, event density, and coding conventions [2]. To mitigate such cross-system discrepancies in our work, we will map all diagnosis codes to the Observational Medical Outcomes Partnership (OMOP) Common Data Model, an open-source relational schema that harmonises disparate coding systems into a single, standardised vocabulary, and apply cluster strategies that preserve consistency across datasets [7].

Methods

Our validation strategy will systematically test cluster robustness across diverse healthcare systems.

A multi-center data collection will include multiple datasets from different populations to test generalization across different groups. Each dataset presents unique challenges in terms of coding systems, completeness, and follow-up duration. For mitigating cross system inaccuracies, we will map ICD-9/ICD-10 codes to standardized clinical concepts defined by the Observational Medical Outcomes Partnership (OMOP) Common Data Model, an open-source relational schema that unifies disparate vocabularies into a single, interoperable framework [7]. This step enables consistent feature extraction and fair comparison across datasets.

The cluster stability analysis will rigorously test whether discovered clusters represent authentic disease subtypes or a random data structure of individual datasets. We will bootstrap resample each cohort multiple times, recording how often the same patients co-cluster.

For cross-dataset validation, after mapping all registries to the OMOP Common Data Model, we will run a simple validation (confounder test): train a logistic regression model that tries to predict registry identity (DNPR vs. US-VA) using only one-hot cluster labels. If the resulting area-under-the-ROC curve (AUC) is ≤ 0.65, we conclude that the OMOP harmonisation has removed most system-specific signal and that clusters capture genuine clinical structure. An AUC > 0.65 indicates that registry differences still dominate, in that case we will revisit OMOP concept mapping before proceeding.

Clinical validity will be assessed by a panel of oncologists who will review cluster summaries and judge whether the trajectories make medical sense across settings. Finally, we will conduct a temporal stability analysis, re-training DEC on non-overlapping calendar periods (e.g., 2000-2010 vs. 2011-2021) to confirm that cluster structure persists over time.

Expected Results

We expect that at least three core clusters will consistently replicate across datasets with an adjusted Rand index (ARI) of ≥ 0.7, suggesting that these represent stable and clinically meaningful phenotypes. These clusters should generalize across healthcare systems despite differences in coding conventions, population demographics, and disease recording practices. The confounder test (AUC ≤ 0.65) will confirm that cluster assignments are not simply artifacts of the source registry. We also anticipate that expert review will validate the clinical plausibility of the dominant clusters, reinforcing their utility for decision-making. Temporal stability analyses are expected to show that key clusters persist over time, supporting their relevance for long-term application in screening and intervention programs.

Translate clusters into actionable, interpretable risk-stratification tools

Relevant Background

Clinical decision-support adoption depends on transparency and actionable recommendations. Sendak et al. proposed “Model Facts” labels summarizing intended use, data set provenance, and key caveats [3]. Earlier work on deep clustering showed that identifying phenotypic subgroups can support personalized predictions [6]. However, it provided limited guidance on which surveillance methods to use and how often to apply them.

Methods

- **Risk-trajectory profiles:** For each cluster, we will first generate Kaplan–Meier survival curves, quantifying how quickly pancreatic cancer emerges after a patient’s index date. We will complement these with smoothed hazard functions to identify time windows in which risk accelerates. Early marker diseases, such as new-onset diabetes or episodes of pancreatitis, will be ranked within each cluster by their mean integrated-gradients attribution score, highlighting which ICD events are most predictive of upcoming transition to high-risk periods.
- **Surveillance mapping:** Building on the risk trajectory analyses, we will develop decision algorithms that recommend when to begin screening, which imaging method to use (MRI or endoscopic ultrasound), and how often to screen (every 6 or 12 months), based on each patient cluster’s risk profile.
  To evaluate their clinical value, we will use decision curve analysis, a method that balances the benefit of correctly identifying high-risk cases (true positives) against the harm of unnecessary follow-ups (false positives). We will compare our tailored strategies to current high-risk screening criteria, such as those based on chronic pancreatitis or family history.

Expected Results

We expect that each cluster will exhibit a distinct risk trajectory, with differences in the timing and intensity of cancer onset as captured by Kaplan–Meier curves and hazard functions. Key early warning signs, such as new-onset diabetes or episodes of pancreatitis, will vary in importance across clusters, as identified by integrated gradients. Based on these patterns, we anticipate developing cluster-specific surveillance strategies that recommend tailored screening intervals and modalities. In decision curve analysis, these personalized strategies are expected to demonstrate greater net clinical benefit than existing one-size-fits-all criteria, by improving early detection while minimizing unnecessary interventions. These tools will provide actionable guidance for clinicians and support real-world implementation of precision screening.

Significance and Conclusions

This research addresses critical gaps in current pancreatic cancer prediction methods by introducing an innovative temporal clustering approach. By developing algorithms capable of capturing complex patterns in disease progression, we aim to move beyond simplistic risk scores towards clinically meaningful, actionable insights. These insights have the potential to enhance early detection, facilitate timely interventions, and ultimately improve patient outcomes.

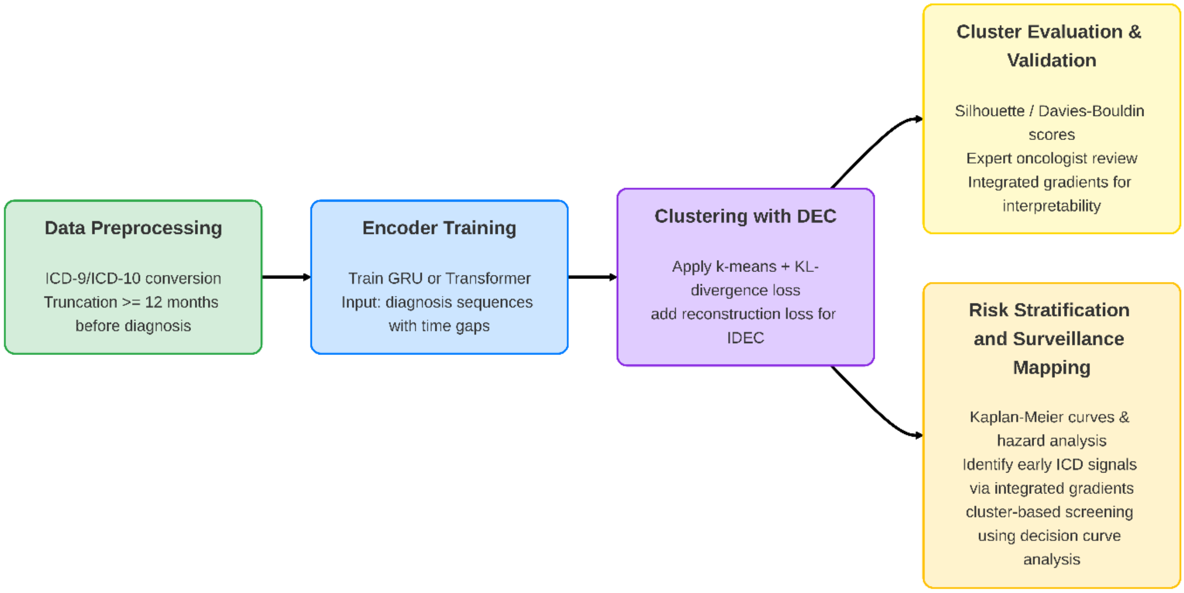
The significance of this work extends well beyond pancreatic cancer alone. The proposed temporal clustering framework could serve as a generalizable model for various diseases characterized by complex, heterogeneous progression trajectories. Conditions such as Alzheimer's disease, heart failure, chronic kidney disease, and even multiple sclerosis similarly involve intricate temporal dynamics that traditional predictive methods fail to capture effectively. By adapting our clustering methodology, we could substantially advance risk prediction and stratification across multiple clinical domains.

By aligning screening strategies with patient-specific disease trajectories, this model enables precision medicine approaches that improve early detection, optimize interventions, and increase both clinical and system-wide efficiency.

Furthermore, from a healthcare systems perspective, improved risk stratification enables more efficient resource allocation. By focusing screening resources and medical attention specifically on patients at highest risk - as accurately identified through temporal clustering - healthcare systems can enhance their cost-effectiveness, reduce unnecessary tests and interventions while improving clinical outcomes for those most in need.

Scientifically, identifying robust and reproducible disease progression clusters may yield valuable biological insights. Persistent clusters across diverse populations and healthcare systems could indicate distinct biological subtypes or mechanistic pathways of pancreatic cancer development. This knowledge would not only inform targeted preventive strategies but also stimulate novel avenues for biological research and therapeutic innovation. Ultimately, this integrative approach could catalyze the development of precision medicine strategies across numerous chronic and life-threatening diseases.

Research Proposal Scheme



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